MSME IDEA HACKATHON 6.0

Reference No. :- **26INC06TN036497**	
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6. Theme	Other Frontier Technologies
7. Idea Sector	Miscellaneous Sector (Environment, Forests, Water & Sanitation; Foods, Beverages, FMCG, Consumer Goods; Infrastructure, Construction, Housing; IT, ITES, Electronics, White Goods, Telecommunication; Metals, Engineering, Machinery, Automation and Transportation, Automotive, E Vehicles, Railways, Aviation, UAV and any other sub-sector)
8. Title of proposed idea/innovation	VanRakshak Autonomous Edge AI Drone Platform for Real-Time Forest Protection and Wildlife Intelligence
9. Briefly explain newness/uniqueness of the innovation	VanRakshak AI is an indigenous autonomous wildlife intelligence platform that combines AI-enabled drones, multi-sensor fusion, edge computing, and autonomous decision-making for continuous forest surveillance. Unlike conventional drones that only capture video for later human review, VanRakshak AI performs real-time onboard threat analysis using RGB, thermal, LiDAR, and audio sensors to detect poaching, illegal logging, forest fires, encroachments, and suspicious activities. The system integrates computer vision, multi-object tracking, mission memory, and a rule-based autonomous decision engine to classify threats and generate GPS-tagged alerts with image and video evidence. This significantly reduces response time while minimizing dependence on continuous manual monitoring. The platform is modular, allowing different mission payloads such as sirens, communication relays, and future fire-response modules. Designed as an indigenous and scalable solution, VanRakshak AI has applications beyond wildlife conservation, including disaster management, border surveillance, critical infrastructure monitoring, and environmental protection.
10. Concept & Objective	The objective of VanRakshak AI is to develop a prototype autonomous aerial surveillance platform capable of assisting Forest Departments in protecting wildlife and natural resources through intelligent monitoring and rapid incident response. The system consists of an autonomous drone equipped with RGB, thermal, LiDAR, GPS, IMU, and audio sensors connected to an onboard Edge AI computing platform. During patrol missions, sensor data is processed locally using computer vision, object detection, tracking, and threat classification algorithms. Based on the detected situation, the onboard decision engine evaluates risk levels and automatically generates GPS-tagged alerts with supporting visual evidence for forest authorities. The proposed prototype aims to demonstrate autonomous patrol, real-time detection of poaching, illegal logging, fire, and unauthorized intrusions, and rapid communication with field personnel through an integrated monitoring platform. The long-term vision is to evolve VanRakshak AI into a scalable Wildlife Intelligence Platform supporting multiple autonomous drones, predictive analytics, and collaborative monitoring for conservation agencies across India.

11. Specify the potential areas of application in industry/market in brief	VanRakshak AI is designed primarily for Forest Departments, Wildlife Protection Authorities, National Parks, Tiger Reserves, Biosphere Reserves, and Environmental Conservation Agencies to improve surveillance, reduce response time, and protect natural resources. Beyond wildlife conservation, the platform can be deployed for border and coastal ecological surveillance, disaster management, forest fire detection, illegal mining monitoring, pipeline and power transmission corridor inspection, railway track surveillance, large industrial campus security, smart city perimeter monitoring, and critical infrastructure protection. The modular architecture enables customization for different mission requirements through interchangeable payloads and AI models, making the platform suitable for both government agencies and private organizations requiring autonomous aerial monitoring solutions.
12. Briefly provide the market potential of idea/innovation	India has an increasing demand for indigenous drone technologies driven by initiatives such as Digital India, Make in India, Drone Rules 2021, and the promotion of domestic UAV manufacturing. Forest Departments across multiple states are modernizing wildlife protection and environmental monitoring through AI-enabled surveillance technologies, creating significant opportunities for scalable indigenous solutions. VanRakshak AI addresses this need by offering an affordable, modular, and intelligent autonomous surveillance platform capable of reducing operational costs, improving monitoring efficiency, and enhancing public safety. The solution follows a scalable business model comprising drone hardware sales, Drone-as-a-Service (DaaS), Annual Maintenance Contracts (AMC), AI analytics subscriptions, and custom deployments. The platform can expand beyond forest surveillance into sectors such as disaster response, infrastructure inspection, environmental monitoring, industrial security, and critical asset protection, providing strong long-term commercialization potential in both government and enterprise markets.
13. Uploaded Block diagram/ flow chart/ Circuit Diagram/Pictures	View/Download